INCH-POUND

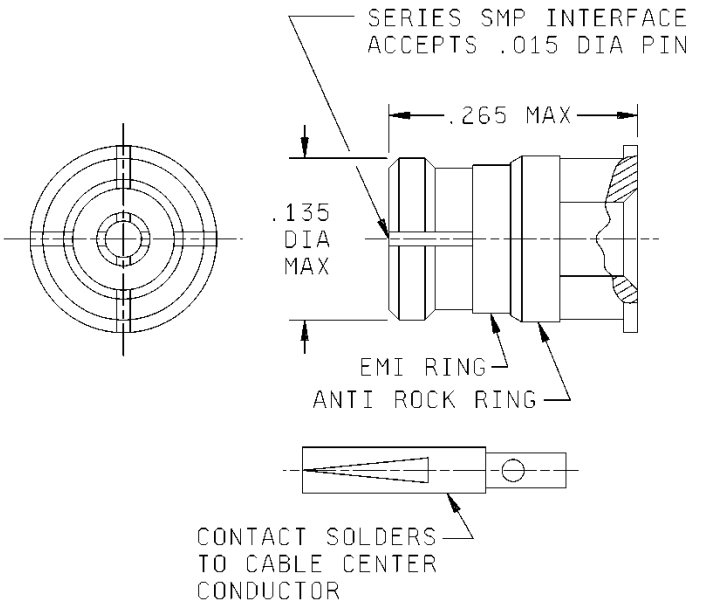
MIL-PRF-31031/31A
w/AMENDMENT 2
16 February 2024
SUPERSEDING
MIL-PRF-31031/31A
w/AMENDMENT 1
23 April 2019

PERFORMANCE SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SOCKET CONTACT, SERIES SMP, FOR .047 AND .086 SEMIRIGID CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-31031.



PIN	Cable accommodation	Cable overall diameter
M31031/31-E4N01	M17/133-RG405 1/	.086 inch
M31031/31-E4N02	M17/151-00001 1/	.047 inch

1/ These connectors accept all cables listed on the cited specification sheet.

NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Special tools shall be required for assembly. Contact the manufacturer.
4. Interface in accordance with MIL-STD-348, series SMP, socket contact.

FIGURE 1. General configuration for -1 and -2.

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ENGINEERING DATA

Nominal impedance: 50 ohms

Frequency range: DC to 40 GHz.

RF leakage: -80 dB to 3 GHz. -65 dB from 3 to 26.5 GHz minimum.

Voltage rating: 335 V rms maximum at sea level: 65 V rms at 70,000 feet.

Operating temperature: -65°C to +165°C.

REQUIREMENTS (after assembly):

Dimensions and configuration: See figure 1.

Mating interface: In accordance with MIL-STD-348.

Force to engage: Prior to performing the following tests a maximum of 3 engagements for conditioning are permitted. Use of the SMP gauges specified in DSCC drawing 06026 is required when performing these tests. The same connector shall not be used for each test. One connector to be used for the full detent gauges, one separate connector to be used with the limited detent gauges and one separate connector is to be tested with the smooth bore gauges. 15 pounds, maximum, full detent shroud, 10 pounds, maximum, limited detent and 2 pounds, maximum, for the smooth bore, in accordance with MIL-PRF-31031/30. When fully engaged the anti rock ring shall perform its intended function.

Force to disengage when using the appropriate gauge and tests in accordance with DSCC drawing 06026: 5 pounds, minimum, full detent shroud, 2 pounds, minimum, limited detent, and .5 pounds, minimum, for smooth bore in accordance with MIL-PRF-31031/30.

Radial misalignment: Not applicable.

Axial misalignment: Not applicable.

Center contact inspection conditions (after heat treat)

Oversize test pin:

Test pin diameter: .0165 inch +.0001/-.0000 inch.

Insertion depth: .065 inch minimum, .070 inch maximum.

Number of insertions: 10.

Insertion force test:

Test pin diameter: .0160 inch +.0001/-.0000 inch.

Insertion force: 24 ounces, maximum.

Insertion depth: .055 inch minimum, .065 inch maximum.

Withdrawal force test:

Test pin diameter: .0140 +0000/-.0001 inch.

Withdrawal force: 0.5 ounce, minimum.

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Insertion depth: .055 inch minimum, .065 inch maximum.

Coupling proof torque: Not applicable.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Center contact retention: 1.5 pounds minimum axial force.

Radial torque: Not applicable.

Voltage standing wave ratio:

DC to 18 gigahertz: 1.20:1 maximum.
18 to 26.5 gigahertz: 1.35:1 maximum.
26.5 to 40 gigahertz: 1.7:1 maximum.

VSWR procedure.

The VSWR shall be measured in accordance with the following procedure or a method acceptable to the Government.

Part should be tested using a Network Analyzer with the Time Domain (TDR) option installed. This is essential to allow the effect of the adapters to be "gated" out. The recommended network analyzer systems include suitable equipment with NIST traceability or equivalent. All gating will be completed in accordance with the procedures stated by the network analyzer manufacturer with the following time domain settings:

TABLE I. Suggested Time Domain Settings.

Transform Mode 1/	Gate Type	Gate Shape 2/	Start Gate Time (ps)	Stop Gate Time (ps)
Low-pass	Bandpass	Minimum, Normal, Wide	End of Adapter 3/ – Rise Time of Gate 4/	End of Device Under Test + Rise Time of Gate

1/ The low-pass mode will have a higher resolution than the band-pass mode.
2/ Each shape specified will not be smaller than the minimum gate span to ensure the response has the correct start and stop times.
3/ "Port Extension" is also used for detection to move the measurement plane after calibration is complete.
4/ The minimum window is used for best resolution.
5/ The maximum window is used for best dynamic range.

The printer/plotter and the computer should be any unit compatible to the system. The test adapters will utilize the following frequency bands:

Frequency	VSWR
500 MHz	1.055
1 GHz	1.055
5 GHz	1.075
10 GHz	1.100
15 GHz	1.125
18 GHz	1.140
26 GHz	1.180

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50-ohm precision air lines may also be used on both sides of the device under test.

Calibration of the system should be performed using the manufacturer's calibration kits and the recommended calibration procedures. The frequency range shall be DC to 26.5 GHz. The VSWR calibration test setup shall be verified using the manufacturers verification kits. The calibrated system VSWR shall be less than 1.02 + .001F (F in GHZ).

The VSWR of the DUT shall be measured using the procedures described in the manufacturer's operating instructions. The time domain shall then be used to remove the effects of the test adapter with the recommended settings as mentioned above.

Contact resistance (in milliohms, maximum):

	Initial
Center contact	6.0
Outer contact	2.0
Outer conductor to connector body	0.5

Dielectric withstanding voltage at sea level: 500 V rms minimum, 125 V rms at 70,000 feet.

Vibration, high frequency (when mated with a full detent shroud in accordance with MIL-PRF-31031/30): MIL-STD-202-204, test condition D.

Vibration, random (when mated with a full detent shroud in accordance with MIL-PRF-31031/30): MIL-STD-202-214, test condition F.

Corona level:

Altitude: 70,000 feet.
190 V rms, minimum.

Insulation resistance: 5,000 megohms minimum when tested in accordance with MIL-PRF-31031.

Altitude: Sea level to 70,000 feet.

Solderability: Applicable.
Shock (specified pulse): MIL-STD-202-213, test condition I. (When mated with full detent shroud in accordance with MIL-PRF-31031/30).

Thermal shock: MIL-STD-202-107, test condition B (except high temperature to be +165°C or maximum high temperature of cable).

RF high potential withstanding voltage:

325 V rms minimum at sea level.
Frequency: 5 MHz.
Leakage current: Not applicable.

Cable retention:

30 pounds minimum: 16 inch pounds torque, applicable (M31031/31-E4N01).
20 pounds minimum: torque, not applicable (M31031/31-E4N02).

Coupling mechanism retention force: Not applicable.

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Durability: 100 cycles.

RF insertion loss: $.12\sqrt{F(GH_z)}$ dB

Part or Identifying Number (PIN): M31031/31-E4N01 and M31031/31-E4N02 (See figure 1).

Supersession: This specification supersedes DSCC drawing 94008 when a QPL source becomes available.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-31031, this document references the following:

MIL-STD-202-107	MIL-STD-348
MIL-STD-202-204	MIL-PRF-31031/30
MIL-STD-202-213	DSCC drawing 06026
MIL-STD-202-214	

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2024-041)

Review activities:
Amy - AT, AV, EA, MI
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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